

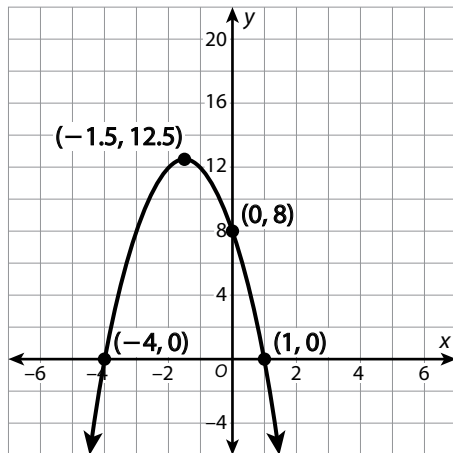


LESSON 21

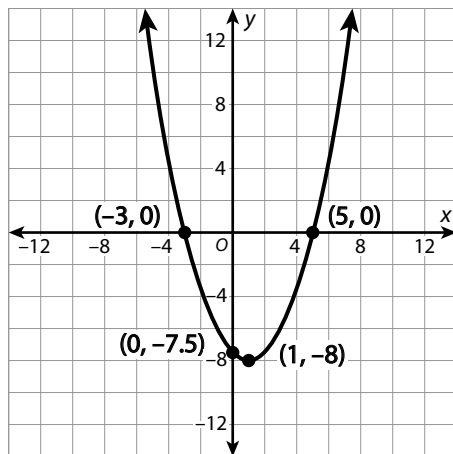
Writing a Quadratic Function from a Graph

- For problems 1 and 2, write an equation in factored form and an equation in vertex form to model the quadratic function given. Show your work.

1



2



- 3 You cannot write an equation for a quadratic function if all you are given are the zeros of the function. Explain.



LESSON 21

Writing a Quadratic Function from a Graph *continued*

- For problems 4–15, write an equation to represent a function f with a graph that passes through the set of points described. Choose from the answers that are mixed up at the bottom of the page. Cross out the answers as you complete the problems.

4 (1, 3)
x-intercepts: -2 and 2

5 (8, 8)
vertex: $(10, 0)$

6 y-intercept: 6
vertex: $(1, 5)$

7 vertex: $(-2, 32)$
zeros: -6 and 2

8 vertex: $(-2, -5)$
y-intercept: -1

9 $(-2, 18)$
vertex: $(0, 10)$

10 vertex: $(-1, 3)$
x-intercepts: -2 and 0

11 $(-1, 8)$
x-intercepts: -5 and 1

12 $(0, 4)$
vertex: $(-3, -5)$

13 $(-1, 0)$
vertex: $(-2, 3)$

14 $(0, 6)$
x-intercepts: 2 and 3

15 $(4, 6)$
vertex: $(6, 4)$

Answers

$$f(x) = (x + 2)^2 - 5$$

$$f(x) = 2(x - 10)^2$$

$$f(x) = (x + 3)^2 - 5$$

$$f(x) = -x^2 - 4x + 5$$

$$f(x) = -3x^2 - 6x$$

$$f(x) = x^2 - 5x + 6$$

$$f(x) = -3(x + 3)(x + 1)$$

$$f(x) = \frac{1}{2}(x - 6)^2 + 4$$

$$f(x) = -2(x + 6)(x - 2)$$

$$f(x) = -(x + 2)(x - 2)$$

$$f(x) = x^2 - 2x + 6$$

$$f(x) = 2(x^2 + 5)$$